

Let's solve these numerical analysis problems step by step using the Bisection Method and Newton-Raphson Method as requested. We'll go through each part systematically, performing up to the specified number of iterations.

Question No. 1

(a) Find the positive root of the equation $xe^x = 1$ which lies between $[0, 1]$ using the Bisection Method (up to 5th iteration)

The Bisection Method requires an interval $[a, b]$ where the function changes sign (i.e., $f(a) \cdot f(b) < 0$). Let's define the function:

$$f(x) = xe^x - 1$$

We need the root in $[0, 1]$.

- Evaluate at the endpoints:

- $f(0) = 0 \cdot e^0 - 1 = 0 - 1 = -1$ (negative)

- $f(1) = 1 \cdot e^1 - 1 = e - 1 \approx 2.718 - 1 = 1.718$ (positive)

Since $f(0) \cdot f(1) < 0$, there's a root in $[0, 1]$. Now, apply the Bisection Method:

• **Iteration 1:**

- Midpoint: $c = \frac{0+1}{2} = 0.5$
- $f(0.5) = 0.5 \cdot e^{0.5} - 1$, where $e^{0.5} \approx 1.6487$, so $0.5 \cdot 1.6487 - 1 \approx 0.82435 - 1 = -0.17565$ (negative)
- Since $f(0.5) < 0$ and $f(1) > 0$, the root is in $[0.5, 1]$.

• **Iteration 2:**

- New interval: $[0.5, 1]$
- Midpoint: $c = \frac{0.5+1}{2} = 0.75$
- $f(0.75) = 0.75 \cdot e^{0.75} - 1$, where $e^{0.75} \approx 2.117$, so $0.75 \cdot 2.117 - 1 \approx 1.58775 - 1 = 0.58775$ (positive)
- Since $f(0.5) < 0$ and $f(0.75) > 0$, the root is in $[0.5, 0.75]$.

• **Iteration 3:**

- New interval: $[0.5, 0.75]$
- Midpoint: $c = \frac{0.5+0.75}{2} = 0.625$
- $f(0.625) = 0.625 \cdot e^{0.625} - 1$, where $e^{0.625} \approx 1.8682$, so $0.625 \cdot 1.8682 - 1 \approx 1.167625 - 1 = 0.167625$ (positive)
- Since $f(0.5) < 0$ and $f(0.625) > 0$, the root is in $[0.5, 0.625]$.

• **Iteration 4:**

- New interval: $[0.5, 0.625]$
- Midpoint: $c = \frac{0.5+0.625}{2} = 0.5625$
- $f(0.5625) = 0.5625 \cdot e^{0.5625} - 1$, where $e^{0.5625} \approx 1.754$, so $0.5625 \cdot 1.754 - 1 \approx 0.986625 - 1 = -0.013375$ (negative)
- Since $f(0.5625) < 0$ and $f(0.625) > 0$, the root is in $[0.5625, 0.625]$.

- **Iteration 5:**

- New interval: $[0.5625, 0.625]$
- Midpoint: $c = \frac{0.5625 + 0.625}{2} = 0.59375$
- $f(0.59375) = 0.59375 \cdot e^{0.59375} - 1$, where $e^{0.59375} \approx 1.811$, so $0.59375 \cdot 1.811 - 1 \approx 1.0753 - 1 = 0.0753$ (positive)
- Since $f(0.5625) < 0$ and $f(0.59375) > 0$, the root is in $[0.5625, 0.59375]$.

After 5 iterations, the approximate root is the midpoint of the final interval:

$$\text{Root} \approx \frac{0.5625 + 0.59375}{2} = 0.578125$$

(b) Use the Bisection Method to find the solution for the equation $2e^x - \sin x = 0$ (up to 5th iteration)

Rewrite the equation as:

$$f(x) = 2e^x - \sin x$$

We need to find an interval where $f(x)$ changes sign. Since no interval is provided, let's test values to find a suitable starting interval. The equation $2e^x = \sin x$ suggests a root near $x = 0$ (as $\sin x$ oscillates between $[-1, 1]$, and e^x grows exponentially).

- Try $x = 0$: $f(0) = 2e^0 - \sin 0 = 2 \cdot 1 - 0 = 2$ (positive)
- Try $x = -1$: $f(-1) = 2e^{-1} - \sin(-1) = 2 \cdot e^{-1} - (-\sin 1) \approx 2 \cdot 0.3679 + \sin 1 \approx 0.7358 + 0.8415 = 1.5773$ (positive)
- Try $x = -2$: $f(-2) = 2e^{-2} - \sin(-2) = 2 \cdot e^{-2} - (-\sin 2) \approx 2 \cdot 0.1353 + \sin 2 \approx 0.2706 + 0.9093 = 1.1799$ (positive)
- Try $x = -3$: $f(-3) = 2e^{-3} - \sin(-3) = 2 \cdot e^{-3} - (-\sin 3) \approx 2 \cdot 0.0498 + \sin 3 \approx 0.0996 + 0.1411 = 0.2407$ (positive)
- Try $x = -4$: $f(-4) = 2e^{-4} - \sin(-4) = 2 \cdot e^{-4} - (-\sin 4) \approx 2 \cdot 0.0183 + \sin 4 \approx 0.0366 - 0.7568 = -0.7202$ (negative)

Now we have an interval $[-4, -3]$ where $f(-4) \cdot f(-3) < 0$. Let's apply the Bisection Method:

• **Iteration 1:**

- Midpoint: $c = \frac{-4 + (-3)}{2} = -3.5$
- $f(-3.5) = 2e^{-3.5} - \sin(-3.5) \approx 2 \cdot 0.0301 + \sin 3.5 \approx 0.0602 - 0.3508 = -0.2906$ (negative)
- Root is in $[-3.5, -3]$.

• **Iteration 2:**

- Midpoint: $c = \frac{-3.5 + (-3)}{2} = -3.25$
- $f(-3.25) = 2e^{-3.25} - \sin(-3.25) \approx 2 \cdot 0.0388 + \sin 3.25 \approx 0.0776 - 0.1087 = -0.0311$ (negative)
- Root is in $[-3.25, -3]$.

• **Iteration 3:**

- Midpoint: $c = \frac{-3.25 + (-3)}{2} = -3.125$
- $f(-3.125) = 2e^{-3.125} - \sin(-3.125) \approx 2 \cdot 0.0438 + \sin 3.125 \approx 0.0876 + 0.0156 = 0.1032$ (positive)
- Root is in $[-3.25, -3.125]$.

• **Iteration 4:**

- Midpoint: $c = \frac{-3.25 + (-3.125)}{2} = -3.1875$
- $f(-3.1875) = 2e^{-3.1875} - \sin(-3.1875) \approx 2 \cdot 0.0413 + \sin 3.1875 \approx 0.0826 - 0.0465 = 0.0361$ (positive)
- Root is in $[-3.25, -3.1875]$.

• **Iteration 5:**

- Midpoint: $c = \frac{-3.25 + (-3.1875)}{2} = -3.21875$
- $f(-3.21875) = 2e^{-3.21875} - \sin(-3.21875) \approx 2 \cdot 0.0400 + \sin 3.21875 \approx 0.0800 - 0.0775 = 0.0025$ (positive)
- Root is in $[-3.25, -3.21875]$.

After 5 iterations, the approximate root is:

$$\text{Root} \approx \frac{-3.25 + (-3.21875)}{2} = -3.234375$$

Question No. 2

(a) Use Newton-Raphson Method to find the root of the equation $x^3 - 2x - 5 = 0$ with initial guess $x_0 = 2.5$ (up to 3rd iteration)

The Newton-Raphson formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Define the function:

$$f(x) = x^3 - 2x - 5$$

Compute the derivative:

$$f'(x) = 3x^2 - 2$$

Start with $x_0 = 2.5$:

- **Iteration 1:**

- $f(2.5) = (2.5)^3 - 2(2.5) - 5 = 15.625 - 5 - 5 = 5.625$
- $f'(2.5) = 3(2.5)^2 - 2 = 3 \cdot 6.25 - 2 = 18.75 - 2 = 16.75$
- $x_1 = 2.5 - \frac{5.625}{16.75} \approx 2.5 - 0.3358 = 2.1642$

- **Iteration 2:**

- $f(2.1642) = (2.1642)^3 - 2(2.1642) - 5 \approx 10.139 - 4.3284 - 5 \approx 0.8106$
- $f'(2.1642) = 3(2.1642)^2 - 2 \approx 3 \cdot 4.6838 - 2 \approx 14.0514 - 2 = 12.0514$
- $x_2 = 2.1642 - \frac{0.8106}{12.0514} \approx 2.1642 - 0.0672 = 2.0970$

- **Iteration 3:**

- $f(2.0970) = (2.0970)^3 - 2(2.0970) - 5 \approx 9.2205 - 4.194 - 5 \approx 0.0265$
- $f'(2.0970) = 3(2.0970)^2 - 2 \approx 3 \cdot 4.397 - 2 \approx 13.191 - 2 = 11.191$
- $x_3 = 2.0970 - \frac{0.0265}{11.191} \approx 2.0970 - 0.0024 = 2.0946$

After 3 iterations, the approximate root is:

$$\text{Root} \approx 2.0946$$

(b) Solve by using Newton-Raphson Method for equation: [Missing]

The problem statement for part (b) of Question 2 is incomplete (“Solve by using Newton-Raphson Method for equation:” with no equation provided). Since no equation is given, I cannot proceed with this part. If you provide the equation, I’d be happy to solve it!

Final Answers

- **Question 1(a):** The positive root of $xe^x = 1$ in $[0, 1]$ after 5 iterations using the Bisection Method is approximately 0.578125.
- **Question 1(b):** The root of $2e^x - \sin x = 0$ after 5 iterations using the Bisection Method is approximately -3.234375 .
- **Question 2(a):** The root of $x^3 - 2x - 5 = 0$ with initial guess $x_0 = 2.5$ after 3 iterations using the Newton-Raphson Method is approximately 2.0946.
- **Question 2(b):** Equation missing; unable to solve.

Let me know if you’d like to provide the missing equation or explore further!